Meet Udeshi

Last updated: Nov. 2026

Ph.D. Candidate, NYU Tandon School of Engineering — Expected graduation: May 2027

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SUMMARY

Ph.D. candidate in CPS security: LLM security agents and benchmarks, binary reverse engineering, malware sandboxing and AI-powered malware. Three years on Qualcomm's AI100 ML compiler.

EDUCATION

2022–present **Ph.D., Electrical Engineering**, NYU Tandon. GPA 3.958/4

2014–2019 **B.Tech. + M.Tech., Electrical Engineering**, IIT Bombay. CGPA 8.18/10

RESEARCH

LLM Agents for Cybersecurity Mar 2024–present

Ph.D. research, Center for Cybersecurity, NYU

Curated NYU CTF Bench (NeurIPS 2024), 200 challenges for evaluating LLM security agents. Developed EnIGMA (ICML 2025), with interactive debugger tools for LLM agents, and the multi-agent D-CIPHER, both state of the art on NYU CTF Bench. Organized the CSAW LLM Attack Challenge.

Cyber-Physical Systems Security Sep 2025–present

Ph.D. research, Control/Robotics Research Lab, NYU

Built an orchestrated QEMU sandbox for CPS malware (EMSOFT 2026) and a sandbox-enabled digital twin (IOLTS 2026). Prototyped *PromptLock*, which ESET initially reported as the first AI-powered ransomware.

Reverse Engineering of Cyber-Physical Systems Aug 2022–present

Ph.D. research, Control/Robotics Research Lab, NYU

Designed REMaQE, symbolic execution that recovers equations from CPS binaries with 100% accuracy (ACM TCPS), and REMEND, a Transformer neural decompiler, 89.8–92.4% accurate on C and Fortran (ACM TIST).

EXPERIENCE

Siemens Healthineers USA May 2026–Aug 2026

AI Cyber Intern, AI Factory

ARPA-H UPGRADE: modeled the cyber-clinical impact of unpatched vulnerabilities in hospitals.

Qualcomm R&D, Bangalore July 2019–July 2022

Senior Engineer, ML Compiler for Cloud AI100

Built multi-core data tiling, memory management and graph scheduling for the AI100 ML compiler.

SELECTED PUBLICATIONS

Equal contributions are marked with †.

[P1] T. Abramovich, **M. Udeshi**†, M. Shao†, et al. "EnIGMA: Interactive Tools Substantially Assist LM Agents in Finding Security Vulnerabilities." *ICML*, 2025.

[P2] M. Shao†, S. Jancheska†, **M. Udeshi**†, B. Dolan-Gavitt†, et al. "NYU CTF Bench: A Scalable Open-Source Benchmark Dataset for Evaluating LLMs in Offensive Security." *NeurIPS*, 2024.

[P3] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. "REMEND: Neural Decompilation for Reverse Engineering Math Equations from Binary Executables." *ACM TIST*, 2025.

[P4] **M. Udeshi**, P. Krishnamurthy, H. Pearce, R. Karri, and F. Khorrami. "REMaQE: Reverse Engineering Math Equations from Executables." *ACM TCPS*, 2024.

HONORS

DAC Young Fellowship (2023), Qualcomm RoCStar award (2021), Indian National Physics Olympiad Gold Medal (2014).

SKILLS

Security	angr, Capstone, Ghidra, Binary Ninja, QEMU, symbolic execution
ML, systems	LLM agents, vLLM, PyTorch, Fairseq, LLVM, Docker, AWS, Slurm
Programming	Python, Embedded C/C++, Shell/Bash, Assembly, Verilog